

1. Run `task_0a.py` to generate the vector database if not already generated.
2. Press Run All (or restart kernel and run all cells).
3. You will be prompted to provide input values.

```
In [ ]: LABEL = input("Provide a valid label name / label ID present in the Caltech101 dataset: ")
        FEATURE_SPACE = input("Provide a feature space [color, hog, avgpool, layer3, layer4, layer5]: ")
        K = int(input("Enter K, the number of similar images K to find for the selected label: "))
```

```
Input label: 0
Downloading dataset if not present.
Files already downloaded and verified
Output label: Faces
```

```
In [ ]: # Load feature space and representative label vectors for that feature space

from utils.database_utils import retrieve

feature_vectors = retrieve(f'{FEATURE_SPACE}.pt')
rep_label_vectors = retrieve(f'rep_label_{FEATURE_SPACE}.pt')
```

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```
for i in range(len(top_k)):

    distance, image_id, label = top_k[i]

    print(str(i + 1) + ".", "\tImage ID: ", image_id, "\t\tDistance: ", distance)
    display(dataset[image_id][0])
```

Downloading dataset if not present.
Files already downloaded and verified

K = 10 most similar images by color feature space using distance: cityblock
1. Image ID: 26 Distance: 11968.653



2. Image ID: 266 Distance: 12236.467



3. Image ID: 82 Distance: 12251.845



4. Image ID: 5058

Distance: 12280.521



5. Image ID: 16

Distance: 12317.934



6. Image ID: 144 Distance: 12624.73



7. Image ID: 22 Distance: 12839.981



8. Image ID: 234 Distance: 13399.064



9. Image ID: 7984 Distance: 13716.107



10. Image ID: 5160

Distance: 13818.619

